CHM2050 Practice Exam B (2nd Mid-Term style) Name: _____

Check Team: □ Argon, □ Barium, □ Cobalt, □ Dysprosium, □ Europium, □ Fluorine, □ Gold.

Answer all questions neatly in the spaces provided. Equations, conversion factors, natural constants, an electronegativity table, and a periodic table are provided in the potentially helpful information at the back of this exam. **Give numerical answers with units and the correct number of significant digits for full credit. Show your work or there will be no points.** The exam consists of fifteen problems with a nominal total of 225 points; it is counted out of 200, so there is an opportunity to earn 25 extra points. The point value and suggested time of each problem are listed. Read all questions first and start with the ones you find easiest. *Total suggested time: 120 min.*

Prob	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	**Σ**
Pts	15	20	10	10	20	10	20	15	10	20	15	20	15	15	10	225
Min	10	10	5	5	10	5	10	10	5	10	5	10	10	10	5	120
Score																

Problem 1 (15 points, 10 min):

The dipole moments and bond lengths of two gas-phase hydrogen halides are: HF, μ = 1.826 D, R = 91.7 pm; HBr, μ = 0.827 D, R = 141.4 pm. (a) Calculate the %-ionic character of each bond. (b) Use the electronegativity table to classify each bond as covalent, polar covalent, or ionic according to the ΔEN rule. Does your result from (a) support the classification of HF? Comment. (c) Draw both molecules with their dipole moment vectors in the convention used in this course.

Problem 2 (20 points, 10 min):

The cyanate ion, OCN^- (C is the central atom), is a stable anion found in many salts, while its isomer, the fulminate ion, CNO^- (N is the central atom), forms shock-sensitive explosive salts such as mercury(II) fulminate. (a) Draw all reasonable Lewis resonance structures of both ions and assign formal charges to every atom. (b) Identify the most important resonance structure of each ion and use formal-charge arguments to explain which isomer should be more stable. (c) Estimate the C–N and C–O bond orders in the cyanate ion by averaging over its two most important resonance structures. (d) Give the shape and the hybridization of the central atom in the cyanate ion.

Problem 3 (10 points, 5 min):

(a) Rank the O–N–O bond angles of NO_2^+, $NO_2^\bullet$, and NO_2^- from largest to smallest and explain your ranking in terms of the space requirements of the electron density regions on N. (b) In SF_4, explain why the lone pair occupies an equatorial and not an axial position, and why the axial F–S–F angle is 173° rather than 180°.

Problem 4 (10 points, 5 min):

Phosphorus pentachloride, PCl_5, is trigonal bipyramidal. Its equatorial P–Cl bonds are 202 pm long while its axial bonds are 214 pm. (a) Give the hybridization of P and state which atomic orbitals of P are used for the equatorial bonds and which for the axial bonds. (b) Explain why the axial bonds are longer. (c) Explain why NCl_5 does not exist even though N is in the same group as P.

Problem 5 (20 points, 10 min):

(a) Use MO theory to explain why He_2 does not exist but the $He_2^{+\bullet}$ cation can be observed in the gas phase. Give both bond orders. (b) Experimentally, B_2 is paramagnetic. Write the electron configuration of B_2 using both possible orderings of the σ_g2p and π_u2p MOs and use the result to decide which ordering is correct for B_2. What effect is responsible for this ordering? (c) Potassium carbide, K_2C_2, contains the C_2^{2-} anion. Give the bond order and magnetism of C_2 and of C_2^{2-}, and rank the C–C bond lengths in C_2, C_2^{2-}, and ethane, C_2H_6. (d) What is the total spin S of ground-state O_2?

Problem 6 (10 points, 5 min):

A 2.500-g sample of a magnesium sulfate hydrate, $MgSO_4 \cdot xH_2O$, is heated until all water of hydration has been driven off. The anhydrous $MgSO_4$ left behind has a mass of 1.221 g. Determine x and the mass-% of water in the hydrate.

Problem 7 (20 points, 10 min):

A 0.2000-g sample of a fragrant liquid that contains only C, H, and O is burned completely in excess O_2. The combustion yields 0.3996 g CO_2 and 0.1636 g H_2O. In a separate experiment, the vapor of the compound has a density of 2.487 g/L at 150. °C and 0.980 atm. (a) Determine the masses of C, H, and O in the sample. (b) Determine the empirical formula. (c) Determine the molar mass from the vapor density and give the molecular formula. (d) Write the balanced combustion equation.

Problem 8 (15 points, 10 min):

Concentrated nitric acid is 70.0% HNO_3 by mass and has a density of 1.413 g/mL. (a) Calculate its molarity, its molality, and the mole fraction of HNO_3. (b) What volume of the concentrated acid is needed to make 500. mL of 1.50 M $HNO_3(aq)$? (c) Name one safety rule for this dilution.

Problem 9 (10 points, 5 min):

200. mL of physiological saline solution (0.154 M NaCl) and 300. mL of 0.100 M $CaCl_2(aq)$ are combined in a 1-L volumetric flask, and the flask is filled to the mark with de-ionized water. (a) Give the concentrations of Na^+, Ca^{2+}, and Cl^- and the total ion concentration in the final solution. (b) What mass of AgCl(s) (143.32 g/mol) would precipitate if excess $AgNO_3$ were added to the whole solution?

Problem 10 (20 points, 10 min):

An antacid tablet containing 0.840 g sodium bicarbonate, $NaHCO_3$ (84.007 g/mol), is dissolved in water and reacts with 25.0 mL of 0.300 M HCl(aq), about the acid in your stomach. (a) Take qualitative chemical inventory of the two solutions before mixing. (b) What types of reaction occur? Write the net ionic equation(s) and identify the conjugate acid/base pairs. (c) Why is HCO_3^- called amphoteric? Give its conjugate acid and conjugate base. (d) Determine the limiting reactant and the volume of $CO_2(g)$ released at body temperature (37 °C) and 1.00 atm. (e) HCl is a strong acid but HF is a weak acid, even though F is more electronegative than Cl. Explain.

Problem 11 (15 points, 5 min):

Ethanol burns according to $C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(g)$. (a) Using the Lewis structure of ethanol and the allocation of bonding electrons by electronegativity, assign the oxidation number of each of the two carbon atoms in ethanol. (b) Give the oxidation numbers of all atoms in O_2, CO_2, and H_2O. (c) Identify the oxidizing and the reducing agent and determine the total number of electrons transferred per molecule of ethanol. (d) Explain why oxidation numbers should not be taken as real charges.

Problem 12 (20 points, 10 min):

A 0.200-g strip of magnesium is dropped into 25.0 mL of 0.800 M HCl(aq). (a) Write the balanced molecular and net ionic equations. (b) Determine the limiting reactant. (c) What volume of dry H_2(g) forms at 25 °C and 755 mm-Hg? (d) After the reaction stops, the remaining acid is titrated with 0.250 M NaOH(aq). What volume of NaOH is needed to reach the equivalence point? (This is called a back-titration.)

Problem 13 (15 points, 10 min):

Uranium for nuclear reactors is enriched by effusion of gaseous UF_6. (a) Calculate the ratio of effusion rates of $^{235}UF_6$ and $^{238}UF_6$ (atomic masses: ^{235}U 235.044 Da, ^{238}U 238.051 Da, F 18.998 Da). (b) Natural uranium contains 0.720% ^{235}U. Each effusion stage multiplies the $^{235}U/^{238}U$ ratio by the factor from (a). How many stages are needed to reach 3.00% ^{235}U? (c) Calculate the root-mean-square speed of $^{238}UF_6$ molecules at 60. °C.

Problem 14 (15 points, 10 min):

A sample of natural gas contains only methane, CH_4, and ethane, C_2H_6. Its density is 0.818 g/L at 25.0 °C and 1.00 atm. (a) Determine the average molar mass of the mixture. (b) Determine the mole fraction of each component. (c) Give the partial pressure of each gas. (d) Which component would deviate more from ideal gas behavior and why?

Problem 15 (10 points, 5 min):

(a) Describe the four colored fields of the NFPA hazard diamond and what the number 4 means in any field. (b) Name the four components that every Standard Operating Procedure (SOP) must contain. (c) In which section of a Safety Data Sheet do you find first-aid measures? (d) Give two specific habits that counteract complacency in a routine laboratory environment.

Potentially Helpful Information (use at your own risk since not all equations are applicable at all times)

Conversion Factors:

$$
\begin{aligned}
1\ \mathrm{kg} &= 2.20462\ \mathrm{lb} \\
1\ \mathrm{lb} &= 16\ \mathrm{oz} \\
1\ \mathrm{mile} &= 1.60934\ \mathrm{km} \\
1\ \mathrm{in} &= 2.54\ \mathrm{cm} \\
1\ \mathrm{ft} &= 0.3048\ \mathrm{m} \\
1\ \mathrm{m} &= 1.09361\ \mathrm{yd} \\
1\ \mathrm{dm^3} &= 1\ \mathrm{L} \\
1\ \mathrm{cm^3} &= 1\ \mathrm{mL} \\
1\ \mathrm{L} &= 1.05669\ \mathrm{qt} \\
1\ \mathrm{Pa} &= 1\ \frac{\mathrm{kg}}{\mathrm{m \cdot s^2}} \\
1\ \mathrm{atm} &= 760\ \mathrm{Torr} = 760\ \mathrm{mm\ Hg} \\
1\ \mathrm{atm} &= 1.01325 \cdot 10^5\ \mathrm{Pa} \\
1\ \mathrm{bar} &= 1 \cdot 10^5\ \mathrm{Pa} \\
1\ \mathrm{J} &= 1\ \frac{\mathrm{kg \cdot m^2}}{\mathrm{s^2}} \\
1\ \mathrm{J} &= 1\ \mathrm{C \cdot V} \\
1\ \mathrm{J} &= 1\ \mathrm{W \cdot s} \\
1\ \mathrm{cal} &= 4.184\ \mathrm{J} \\
1\ \mathrm{atm \cdot L} &= 1.01325 \cdot 10^2\ \mathrm{J} \\
1\ \mathrm{eV} &= 1.602176634 \cdot 10^{-19}\ \mathrm{J} \\
0\ ^\circ\mathrm{C} &= 273.15\ \mathrm{K} \\
1\ \mathrm{ppm} &= 1 \cdot 10^{-6} \\
1\ \text{Å} &= 1 \cdot 10^{-10}\ \mathrm{m} \\
1\ \mathrm{Ry} &= 13.605698\ \mathrm{eV} \\
1\ \mathrm{Da} &= 1\ \mathrm{u} = 1\ \mathrm{g/mol}
\end{aligned}
$$

Natural Constants (†indicates an exact number):

$$
\begin{aligned}
N_\mathrm{A}^\dagger &= 6.02214076 \cdot 10^{23}\ \mathrm{mol^{-1}} \\
e^\dagger &= 1.602176634 \cdot 10^{-19}\ \mathrm{C} \\
F^\dagger &= 9.648533212 \cdot 10^4\ \frac{\mathrm{C}}{\mathrm{mol}} \\
m_e &= 9.1093837015 \cdot 10^{-31}\ \mathrm{kg} \\
m_p &= 1.67262192369 \cdot 10^{-27}\ \mathrm{kg} \\
m_n &= 1.67492749804 \cdot 10^{-27}\ \mathrm{kg} \\
h^\dagger &= 6.62607015 \cdot 10^{-34}\ \mathrm{J \cdot s} \\
\hbar &= h/(2\pi) = 1.054571817 \cdot 10^{-34}\ \mathrm{J \cdot s} \\
c^\dagger &= 2.99792458 \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}} \\
g^\dagger &= 9.80665\ \frac{\mathrm{m}}{\mathrm{s^2}} \\
R^\dagger &= 8.314462618\ \frac{\mathrm{J}}{\mathrm{K \cdot mol}} \\
R &= 8.205736 \cdot 10^{-2}\ \frac{\mathrm{L \cdot atm}}{\mathrm{K \cdot mol}} \\
k_\mathrm{B}^\dagger &= 1.380649 \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}} \\
\varepsilon_0 &= 8.8541878128 \cdot 10^{-12}\ \frac{\mathrm{As}}{\mathrm{Vm}} \\
R_\mathrm{H} &= 1.096775834 \cdot 10^7\ \mathrm{m^{-1}} \\
R_\infty &= 1.0973731568160 \cdot 10^7\ \mathrm{m^{-1}}
\end{aligned}
$$

Equations:

$$
\begin{aligned}
V_\mathrm{cube} &= \ell^3 \\
V_\mathrm{cylinder} &= \left(\pi r^2\right) \cdot \ell \\
V_\mathrm{sphere} &= \frac{4}{3}\pi r^3 \\
T/[\mathrm{K}] &= T/[^\circ\mathrm{C}] + 273.15 \\
T/[^\circ\mathrm{C}] &= (T/[^\circ\mathrm{F}] - 32) \cdot \frac{5}{9} \\
F(r) &= \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} \\
V(r) &= \frac{q_1 q_2}{4\pi\varepsilon_0 r} \\
V_\mathrm{eff} &= -\frac{Z_\mathrm{eff} e^2}{4\pi\varepsilon_0 r} \\
T &= \frac{1}{2}mv^2 \\
p &= mv \\
L &= mvr \\
T &= -\frac{V}{2} \\
\hat{\mathcal{H}} &= \hat{\mathcal{T}} + \hat{\mathcal{V}} \\
\nu &= \frac{c}{\lambda} \\
\vec{E} &= \vec{E}_0 \sin\left\{\vec{k} \cdot \vec{z} + \omega t + \Phi\right\} \\
\left|\vec{k}\right| &= \frac{1}{\lambda} \\
\omega &= 2\pi\nu \\
2\ell &= n\lambda \\
E_n &= nh\nu \\
E_\mathrm{photon} &= h\nu = \frac{hc}{\lambda} \\
L_n &= n\hbar = n\frac{h}{2\pi} \\
E_n &= -\frac{hcR_\mathrm{H}}{n^2} \\
h\nu &= E_{n_2} - E_{n_1} \\
\nu\left(n_2 \longrightarrow n_1\right) &= cR_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \\
\bar{\nu}\left(n_2 \longrightarrow n_1\right) &= R_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \\
\lambda\left(n_2 \longrightarrow n_1\right) &= \frac{1}{R_\mathrm{H}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)} \\
R_\infty &= \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \\
R_\mathrm{H} &= \frac{\mu_H e^4}{8\varepsilon_0^2 h^3 c} \\
\mu &= \frac{m_1 \cdot m_2}{m_1 + m_2} \\
\lambda &= \frac{h}{p} \\
n\lambda &= 2\pi \cdot r \\
\Delta x \cdot \Delta p_x &\leq \frac{\hbar}{2} \\
\Delta\phi \cdot \Delta L &\leq \frac{\hbar}{2} \\
\Delta E \cdot \Delta t &\leq \frac{\hbar}{2}
\end{aligned}
$$

$$\hat{\mathcal{H}} \cdot \psi(x,y,z) = E \cdot \psi(x,y,z)$$
$$\hat{\mathcal{H}} \cdot \psi(r,\theta,\phi) = E \cdot \psi(r,\theta,\phi)$$
$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$
$$\psi(r,\theta,\phi) = R_{nl}(r) \cdot \Theta_{lm_l}(\theta) \cdot \Phi_{m_l}(\phi)$$
$$n = 1, 2, \ldots$$
$$l = 0, 1, 2, \ldots, n-1$$
$$m_l = 0, \pm 1, \pm 2, \ldots, \pm l$$
$$m_s = \pm\frac{1}{2}$$
$$E_n = \frac{-hcR_{\mathrm{H}}}{n^2} = -\frac{\mu e^4}{8\varepsilon_0^2 h^2 n^2}$$
$$R_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c}$$
$$R_{\mathrm{H}} = \frac{\mu e^4}{8\varepsilon_0^2 h^3 c}$$
$$L = I \cdot \omega$$
$$\left|\vec{L}\right| = \sqrt{l(l+1)}\,\hbar$$
$$\left|\vec{S}\right| = \sqrt{s(s+1)}\,\hbar$$
$$\vec{\mu} = \frac{-g_e e}{2m_e}\vec{S}$$
$$\left|\vec{L}_z\right| = |m_l \cdot \hbar|$$
$$\left|\vec{S}_z\right| = |m_s \cdot \hbar|$$
$$L_z = \left|\sum_i m_l(i)\right|$$
$$S = \left|\sum_i m_s(i)\right|$$
$$\mathbf{mole{-}\%}(i) = \frac{n_i}{\sum\limits_i n_i}$$
$$\mathbf{mass{-}\%}(i) = \frac{m_i}{\sum\limits_i m_i}$$
$$\overline{\mathcal{M}} = \sum_i c_i \cdot \mathcal{M}_i$$
$$1 = \sum_i c_i$$
$$n = \frac{m}{\mathcal{M}}$$
$$\text{bond order} = \frac{1}{2}\left(\begin{array}{c}\text{\# of } e^-\text{'s in}\\ \text{bonding MOs}\end{array} - \begin{array}{c}\text{\# of } e^-\text{'s in}\\ \text{antibonding MOs}\end{array}\right)$$
$$\text{bond order} = \left(\begin{array}{c}\text{average \# of bonds}\\ \text{averaged over all}\\ \text{resonance structures}\end{array}\right)$$

$$d = \frac{m}{V}$$
$$n = c_{\mathrm{init}} \cdot V_{\mathrm{init}} = c_{\mathrm{fin}} \cdot V_{\mathrm{fin}}$$
$$M = \frac{n_{\mathrm{solute}}}{V_{\mathrm{solution}}}$$
$$b = \frac{n_{\mathrm{solute}}}{m_{\mathrm{solvent}}}$$
$$\chi = \frac{n_{\mathrm{solute}}}{n_{\mathrm{total}}}$$
$$M \approx d_{\mathrm{solvent}} \cdot b$$
$$n_{\mathrm{ions}} = i \cdot n_{\mathrm{formula\ unit}}$$
$$\%\text{-yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \cdot 100\%$$
$$PV = nRT$$
$$P_{\mathrm{total}} = \sum_i P_i$$
$$Z = \frac{PV}{nRT}$$
$$w = F \cdot d$$
$$F = \frac{P}{A}$$
$$E_{\mathrm{grav}} = mg\Delta h$$
$$E_{\mathrm{kin}} = \frac{m}{2}v^2$$
$$\langle E_{\mathrm{kin}}\rangle = \frac{3}{2}k_{\mathrm{B}}T$$
$$v_{\mathrm{mp}} = \sqrt{\frac{2\,R\,T}{\mathcal{M}}}$$
$$\langle v\rangle = \sqrt{\frac{8\,R\,T}{\pi\mathcal{M}}}$$
$$v_{\mathrm{rms}} = \sqrt{\frac{3\,R\,T}{\mathcal{M}}}$$
$$w = -P_{\mathrm{ext}}\,\Delta V$$
$$q = C \cdot \Delta T$$
$$C = c \cdot m = \overline{C} \cdot n$$
$$\Delta U = q + w$$
$$H = U + PV$$
$$\Delta H = \Delta U + \Delta(PV)$$
$$U = \frac{3}{2}nRT$$
$$\Delta_{\mathrm{rxn}}\overline{H}^\circ = \sum_{\mathrm{products}} \Delta_{\mathrm{f}}\overline{H}^\circ - \sum_{\mathrm{reactants}} \Delta_{\mathrm{f}}\overline{H}^\circ$$

Considerations for Z_{eff} and E_i calculations (Slater's Rules):

$$E_i \approx (-1312\ \mathrm{kJ/mol})\left(\frac{Z_{\mathrm{eff}}(i)}{n_i}\right)^2$$

group	other electrons in same group	electrons in group w/ same n and $< l$	electrons in group w/ $n-1$	electrons in group $< (n-1)$
$[1s]$	0.30	-	-	-
$[ns, np]$	0.35	-	0.85	1
$[nd]$ or $[nf]$	0.35	1	1	1

Pauling Electronegativities

H 2.20																	He –
Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne –
Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar –
K 0.82	Ca 1.0	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.0
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.6
Cs 0.79	Ba 0.89	Lu 1.27	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.0	Tl 1.62	Pb 2.33	Bi 2.02	Po 2.0	At 2.2	Rn –
Fr >0.79	Ra 0.9	Lr –	Rf –	Db –	Sg –	Bh –	Hs –	Mt –	Ds –	Rg –	Cn –	Nh –	Fl –	Mc –	Lv –	Ts –	Og –

La 1.10	Ce 1.12	Pr 1.13	Nd 1.14	Pm –	Sm 1.17	Eu –	Gd 1.20	Tb –	Dy 1.22	Ho 1.23	Er 1.24	Tm 1.25	Yb –
Ac 1.1	Th 1.3	Pa 1.5	U 1.38	Np 1.36	Pu 1.28	Am 1.3	Cm 1.3	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.3

<u>Selected Brønsted-Lowry Acids and Bases:</u>
Strong Acids: $H_2S_2O_7$, HI, HBr, HCl, H_2SO_4, $HClO_4$, HNO_3, $H_3O^+(aq)$.
Weak Acids: $HSO_4^-(aq)$, $H_2C_2O_4(aq)$, $H_2CrO_4(aq)$, $Co(OH_2)_6^{3+}(aq)$, $H_3PO_4(aq)$, $Fe(OH_2)_6^{3+}(aq)$, $H_3AsO_4(aq)$, HF(aq), $HC_2O_4^-(aq)$, $CH_3COOH(aq)$, $Al(OH_2)_6^{3+}(aq)$, $H_2CO_3(aq)$.,$NH_4^+(aq)$, $H_2S(aq)$ HCN(aq).
Strong Bases: O^{2-}, S^{2-}, NH_2^-, $OH^-(aq)$.
Weak Bases: $PO_4^{3-}(aq)$, $CO_3^{2-}(aq)$, $NH_3(aq)$, $CN^-(aq)$, $HPO_4^{2-}(aq)$, $HS^-(aq)$, $HAsO_4^{2-}(aq)$, $CrO_4^{2-}(aq)$, $HCO_3^{2-}(aq)$, $CH_3COO^-(aq)$, $C_2O_4^{2-}(aq)$, $F^-(aq)$, $SO_4^{2-}(aq)$.

<u>Selected Polyatomic Ions:</u>

Name	Formula	Name	Formula	Name	Formula
ammonium	NH_4^+	bicarbonate	HCO_3^-	phosphate	PO_4^{3-}
hydronium	H_3O^+	nitrate	NO_3^-	hydrogen phosphate	HPO_4^{2-}
peroxide	O_2^{2-}	nitrite	NO_2^-	dihydrogen phosphate	$H_2PO_4^-$
hydroxide	OH^-	sulfate	SO_4^{2-}	perchlorate	ClO_4^-
acetate	CH_3COO^-	bisulfate	HSO_4^-	chlorate	ClO_3^-
cyanide	CN^-	sulfite	SO_3^{2-}	chlorite	ClO_2^-
azide	N_3^-	bisulfite	HSO_3^-	hypochlorite	ClO^-
carbonate	CO_3^{2-}	hydrogen sulfide	HS^-	permanganate	MnO_4^-

IUPAC Periodic Table of the Elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H hydrogen 1.008 [1.0078, 1.0082]																	2 He helium 4.0026
3 Li lithium 6.94 [6.938, 6.997]	4 Be beryllium 9.0122											5 B boron 10.81 [10.806, 10.821]	6 C carbon 12.011 [12.009, 12.012]	7 N nitrogen 14.007 [14.006, 14.008]	8 O oxygen 15.999 [15.999, 16.000]	9 F fluorine 18.998	10 Ne neon 20.180
11 Na sodium 22.990	12 Mg magnesium 24.305 [24.304, 24.307]											13 Al aluminium 26.982	14 Si silicon 28.085 [28.084, 28.086]	15 P phosphorus 30.974	16 S sulfur 32.06 [32.059, 32.076]	17 Cl chlorine 35.45 [35.446, 35.457]	18 Ar argon 39.95 [39.792, 39.963]
19 K potassium 39.098	20 Ca calcium 40.078(4)	21 Sc scandium 44.956	22 Ti titanium 47.867	23 V vanadium 50.942	24 Cr chromium 51.996	25 Mn manganese 54.938	26 Fe iron 55.845(2)	27 Co cobalt 58.933	28 Ni nickel 58.693	29 Cu copper 63.546(3)	30 Zn zinc 65.38(2)	31 Ga gallium 69.723	32 Ge germanium 72.630(8)	33 As arsenic 74.922	34 Se selenium 78.971(8)	35 Br bromine 79.904 [79.901, 79.907]	36 Kr krypton 83.798(2)
37 Rb rubidium 85.468	38 Sr strontium 87.62	39 Y yttrium 88.906	40 Zr zirconium 91.224(2)	41 Nb niobium 92.906	42 Mo molybdenum 95.95	43 Tc technetium	44 Ru ruthenium 101.07(2)	45 Rh rhodium 102.91	46 Pd palladium 106.42	47 Ag silver 107.87	48 Cd cadmium 112.41	49 In indium 114.82	50 Sn tin 118.71	51 Sb antimony 121.76	52 Te tellurium 127.60(3)	53 I iodine 126.90	54 Xe xenon 131.29
55 Cs caesium 132.91	56 Ba barium 137.33	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59	81 Tl thallium 204.38 [204.38, 204.39]	82 Pb lead 207.2	83 Bi bismuth 208.98	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson

Key:

atomic number
Symbol
name
conventional atomic weight
standard atomic weight

57 La lanthanum 138.91	58 Ce cerium 140.12	59 Pr praseodymium 140.91	60 Nd neodymium 144.24	61 Pm promethium	62 Sm samarium 150.36(2)	63 Eu europium 151.96	64 Gd gadolinium 157.25(3)	65 Tb terbium 158.93	66 Dy dysprosium 162.50	67 Ho holmium 164.93	68 Er erbium 167.26	69 Tm thulium 168.93	70 Yb ytterbium 173.05	71 Lu lutetium 174.97
89 Ac actinium	90 Th thorium 232.04	91 Pa protactinium 231.04	92 U uranium 238.03	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

INTERNATIONAL UNION OF PURE AND APPLIED CHEMISTRY

For notes and updates to this table, see www.iupac.org. This version is dated 1 December 2018.

Copyright © 2018 IUPAC, the International Union of Pure and Applied Chemistry.

United Nations Educational, Scientific and Cultural Organization

2019 IYPT International Year of the Periodic Table of Chemical Elements

IUPAC